

Session 2: Security goals

Slides prepared by UMBC Protocol Analysis Lab

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Common Security Goals

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Common Security Goals

- ▶ **Confidentiality:** In a two-party communication protocol, confidentiality holds if no third party is able to access the information sent.

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Common Security Goals

- ▶ **Confidentiality:** In a two-party communication protocol, confidentiality holds if no third party is able to access the information sent.
- ▶ **Agreement:** In a two-party communication protocol, agreement holds if both parties agree on some set of data values in the protocol. These values are then used to derive the key.

Other common security goals

1. Aliveness
2. Weak Agreement
3. Non-injective Agreement
4. A specific relay party cannot read the message
5. The message must be timestamped
6. Perfect forward secrecy
7. Crypto primitives are resistant to a predefined attack

Needham Schroeder

Designing a meeting protocol, and beyond

Strand spaces

A *strand space* is a formalism used to determine the security of a network protocol

- ▶ Fábrega, F. J. T., Herzog, J. C., & Guttman, J. D. (1998, May). [Strand spaces: Why is a security protocol correct?](#)
- ▶ Represent honest communicants and intruders as “strands”
- ▶ Parallel / sequential composability.

Example of a strand

Alice

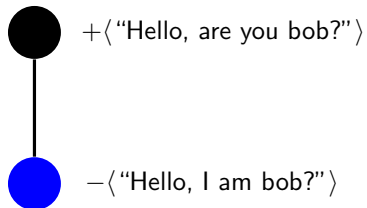
Example of a strand

Alice

 + $\langle \text{"Hello, are you bob?"} \rangle$

Example of a strand

Alice



Strand space

Alice

$+\langle \text{"Hello, are you bob?"} \rangle$ ●

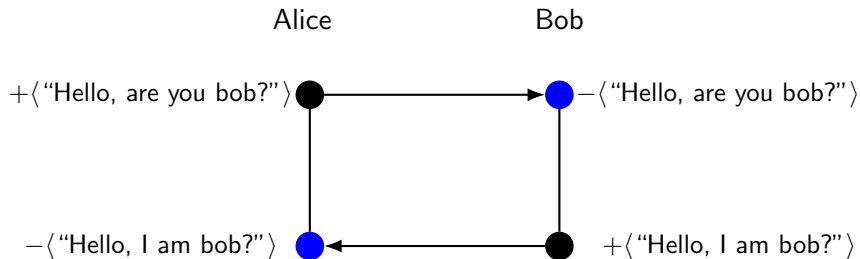
● $-\langle \text{"Hello, I am bob?"} \rangle$

Bob

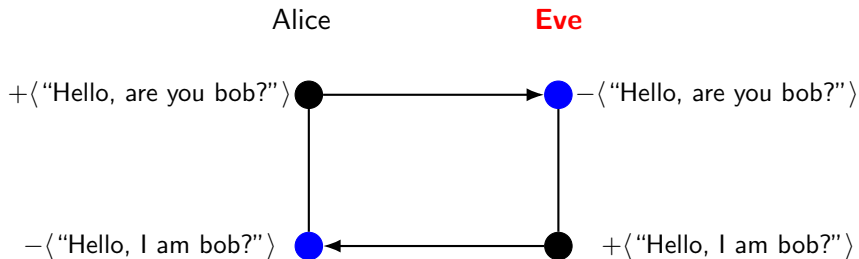
● $-\langle \text{"Hello, are you bob?"} \rangle$

● $+\langle \text{"Hello, I am bob?"} \rangle$

A completed protocol?



Strands can lie



Needham Schroeder

Alice

Bob

● = Reception ● = Transmission

Needham Schroeder

Alice

Bob



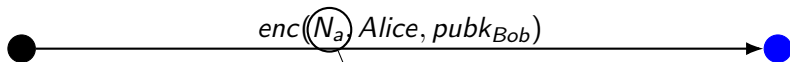
● = Reception

● = Transmission

Needham Schroeder

Alice

Bob



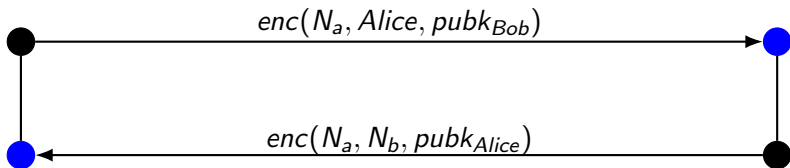
nonce: a *fresh*, cryptographic random number generated by Alice.

● = Reception ● = Transmission

Needham Schroeder

Alice

Bob

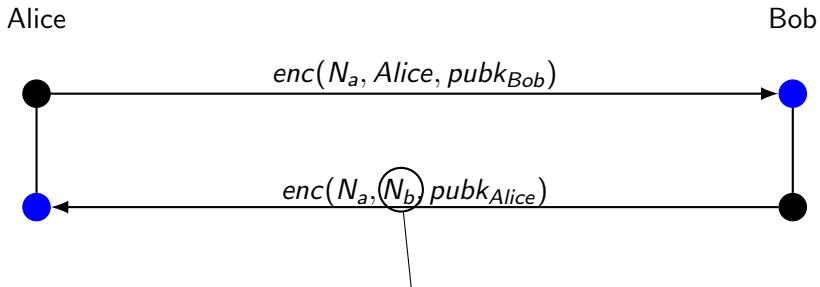


= Reception



= Transmission

Needham Schroeder



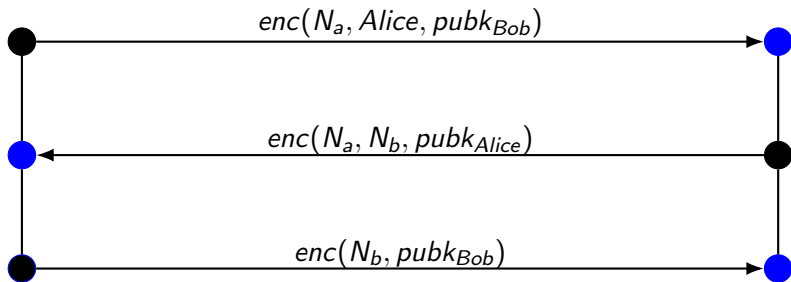
nonce: a *fresh*, cryptographic random number generated by Bob.

● = Reception ● = Transmission

Needham Schroeder

Alice

Bob

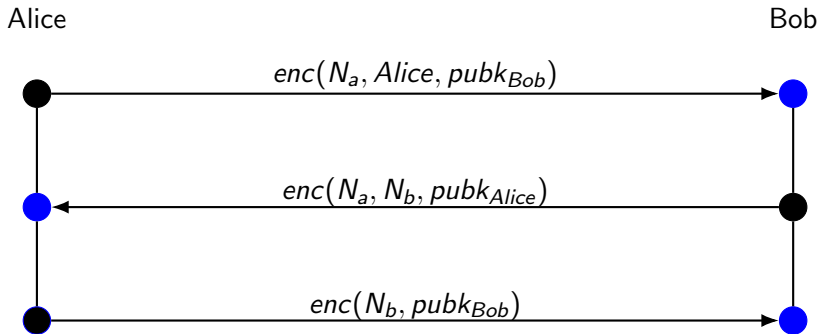


= Reception



= Transmission

Needham Schroeder



Is this protocol secure against the Dolev-Yao Adversary?

Blue circle = Reception Black circle = Transmission

Cryptographic Protocol Shapes Analyzer

- ▶ Tool for analyzing and developing secure protocols.
- ▶ Enumerates all possible executions of a protocol.
- ▶ Assumes a Dolev-Yao adversary.
- ▶ Developed by MITRE, available at:
<https://github.com/mitre/cpsa>

Analyze Needham-Schroeder

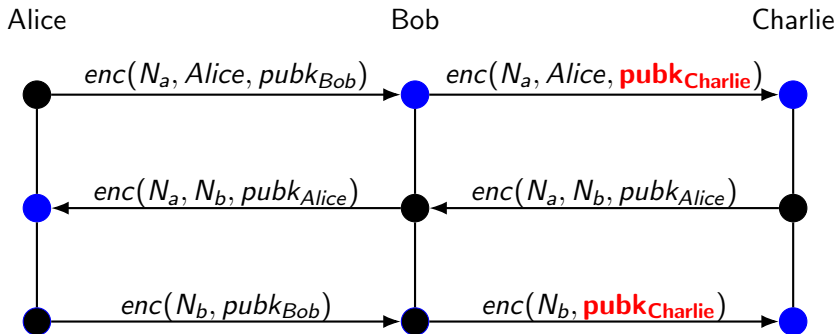
Define protocol *roles* and *skeletons*.

Run CPSA to produce *shapes*.

Analyze *shapes* for expected and unexpected behavior.

Needham Schroeder Attack

Needham Schroeder Attack



Remember: The Dolev-Yao Adversary can be a legitimate party on the network

● = Reception ● = Transmission

Fix your protocol

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Last time, you created a meeting protocol.

Do you think your meeting protocol withstands the Dolev-Yao adversary?

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Do you think your meeting protocol withstands the Dolev-Yao adversary?

Model your protocol in CPSA and see if there is an attack!

See if you can fix your protocol so that there is no more attack.

Survey

Fill in details about the survey here!